# Bilinear Realisations of Truncated Chen–Fliess Series and Path Signatures: a Tutorial

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**Abstract**

This tutorial shows how to turn a truncated Chen–Fliess series, or a truncated path signature, into a small bilinear state-space model, and how to fit such a series to sampled data. It covers systems with a letter for time (a drift) and systems without one, where the series is a path signature. The methods are standard: iterated integrals, the Hankel matrix indexed by words, minimal and graded realisations, Volterra kernels and the Lie–Hankel matrix built from Lyndon words. The tutorial works through them on a synthetic example and four test systems, with code that reproduces every number. It states the practical lessons learnt in applying them. These cover how to compute the iterated integrals from formulas and from samples, and how to set up the Hankel matrix of a truncated series. They cover why each window needs its own starting value, and why words should be selected rather than all fitted with a ridge penalty. They cover what the fitted coefficients can and cannot tell, and why the sizes of realisations of fitted series must be compared with their largest possible values. Finally, they cover how far a realisation can be trusted beyond the window it was fitted on.

## 1 Introduction

Many systems are studied through the way a measured output responds to an input, rather than through known equations. For a linear system, one function describes this response: the impulse response. For a nonlinear system, Volterra replaced it by a hierarchy of kernels, one for each order of interaction between inputs at different times [33]; Brockett connected these kernels to the geometry of the system [2]. The kernels are functions of many time arguments. They are hard to estimate from data, and they do not by themselves give a model with a finite state [34].

Fliess reorganised the same information [9], building on Chen's iterated integrals [3]. The Chen–Fliess series attaches one number to each word: a finite sequence drawn from an alphabet with one letter for the passage of time and one letter for each input. When there is no letter for time, the iterated integrals are the path signature of the inputs, now used widely for time series [22, 11, 5]. Work on path signatures often adds time to the channels as an extra coordinate; the iterated integrals of that augmented path are exactly those of a Chen–Fliess series with a letter for time.

The series has a realisation theory that mirrors linear systems theory. A linear impulse response can be reproduced by a finite linear state-space model, found from its Hankel matrix by the Ho–Kalman method [20, 17]. In the same way, a Chen–Fliess series can be reproduced by a finite bilinear model, found from a Hankel matrix indexed by words [8, 29, 18, 7]. Hespel and Jacob developed this into numerical methods [13, 14]. The link between finite series and nilpotent structure is old. Crouch showed that a finite Volterra series, polynomial in the input, has a realisation whose Lie algebra is nilpotent [6]; that condition concerns brackets of the vector fields, not products of matrices. Sontag studied polynomial response maps [31], and Krener the approximation of nonlinear systems by bilinear ones to a given order [21]. Shaikin used nilpotency of the Lie algebra of the input matrices, with the drift excluded, to obtain finite filters for stochastic bilinear systems [30].

In practice only a truncated part of the series is known, estimated up to some word length $K$ from a finite record of data. This tutorial collects, in one place and with code, what is needed to realise such a series and to fit it. Section 2 introduces words, iterated integrals and the series, and how to compute the integrals. Section 3 introduces bilinear realisations, the Hankel matrix of a truncated series, and two ways of building a realisation. Section 4 works through four systems with exact coefficients: the driven Duffing oscillator, a Duffing-like system without a letter for time, the Heisenberg integrator, whose series is finite, and a system with four input channels. Section 5 fits truncated series to sampled data, Section 6 derives Volterra kernels from the coefficients, and Section 7 introduces Lyndon words and the Lie–Hankel matrix. Section 8 lists the practical lessons.

# 2 Background

## 2.1 Letters, words and iterated integrals

Consider a system with state $x$, input $u(t)$ and output $y$,

$$\dot{x} = f_0(x) + \sum_{i=1}^{m} f_i(x)\, u_i(t), \qquad y = h(x), \qquad x(0) = x_0. \tag{1}$$

The vector field $f_0$ moves the state when there is no input; it is the drift. Each $f_i$ is the direction in which input $i$ pushes the state.

The alphabet has one letter for each of these. Letter 0 stands for the drift, and letters $1, \dots, m$ for the inputs. A word is a finite sequence of letters, $w = (a_1, \dots, a_L)$, read in time order: $a_1$ acts first. Its length is $|w| = L$, and $()$ is the empty word. To each word belongs an iterated integral,

$$I_{()}(t) = 1, \qquad I_{(a_1,\dots,a_L)}(t) = \int_0^t I_{(a_1,\dots,a_{L-1})}(s)\, u_{a_L}(s)\, ds, \qquad u_0 \equiv 1. \tag{2}$$

Each letter integrates the previous integral against its own signal: the drift letter against time, an input letter against that input.

*Example* 1 (The first iterated integrals, one input). With one input ($m = 1$), the integrals of the words of length 1 and 2 are

$$I_{(0)}(t) = t, \qquad I_{(1)}(t) = \int_0^t u(s)\, ds,$$

$$I_{(0,0)}(t) = \tfrac{t^2}{2}, \qquad I_{(0,1)}(t) = \int_0^t s\, u(s)\, ds,$$

$$I_{(1,0)}(t) = \int_0^t (t-s)\, u(s)\, ds, \qquad I_{(1,1)}(t) = \tfrac{1}{2}\Big(\int_0^t u(s)\, ds\Big)^2.$$

A word of drift letters alone is a power of time. The words $(0,1)$ and $(1,0)$ contain the same letters in a different order, and their integrals weight the input differently: early input counts most in $I_{(1,0)}$, late input in $I_{(0,1)}$.

The integral of a word of length $L$ is scaled by $t^L/L!$. Over a short window the integrals of long words are therefore very small. This matters for fitting (Section 5).

When the system has no drift, $f_0 = 0$, there is no letter for time. The letters are the input channels alone, and the iterated integrals are the signature of the path $\xi(t) = \int_0^t u(s)\, ds$ [3, 22]. With two channels, for example, $I_{(1,2)}(t) = \int_0^t \xi_1(s)\, u_2(s)\, ds$. The state then moves only when the path moves: with no input, nothing changes. Without a drift the integral of a word of length $L$ is scaled by $(Ut)^L/L!$, where $U$ is the size of the inputs, because every letter integrates against an input. The rest of this section, and the realisation theory that follows, apply unchanged; only the alphabet differs.

### 2.2 Computing the iterated integrals

The integrals are computed in one of two ways, according to what is known about the input. When the input is known as a formula, they are the solution of the linear equations $\dot{I}_{wa} = u_a I_w$ (with $u_0 = 1$ for the drift letter), integrated by the eighth-order Dormand–Prince method [12] with the input evaluated exactly. Each integral is integrated in units of its own scale, so that the tolerance of the integrator is a relative one even for the tiny integrals of long words. For words of input letters alone and of drift letters alone, which have closed forms, the largest relative error up to length 8 is $1.5 \times 10^{-13}$. When the input is known only as samples, as with recorded data and in every fit of Section 5, the integrals are computed by the iisignature library [27] on the sampled path: $(t, \int_0^t u)$ with a letter for time, $\int_0^t u$ without. The integral of $u$ at the samples is taken as the exact integral of a cubic spline through the samples of $u$, and iisignature gives the exact iterated integrals of the path that joins those points by straight lines. Its difference from the exact integrals is of second order in the sampling step. For the input $u(t) = 0.3 \sin 6\pi t + 0.1 \sin 14\pi t$, sampled every 1 ms over 0.15 s, the error of each integral is measured as its largest difference from the exact value over the window, divided by that integral's own largest size. The worst such relative error over all words up to length 6 is $2.2 \times 10^{-4}$ at a step of 1 ms and $2.2 \times 10^{-6}$ at a step of 0.1 ms. It grows with word length: $1.2 \times 10^{-5}$ for words of length 2 and $2.2 \times 10^{-4}$ for length 6 at 1 ms. For words of length 2 to 6 it falls a hundredfold for a tenfold smaller step, as expected for straight-line joins; for length 1, the integral of a cubic spline, it falls faster. For the faster inputs of Section 5 the error of each integral, relative to its own size over the window, has a median of $1.7 \times 10^{-5}$ at length 2 and $4.7 \times 10^{-4}$ at length 6; it reaches $2.5 \times 10^{-4}$ and $1.7 \times 10^{-2}$ for integrals whose largest value is very small.

### 2.3 The Chen–Fliess series

The output of (1) is

$$y(t) = \sum_w c_w \, I_w(t), \qquad c_{(a_1,\dots,a_L)} = \big(L_{f_{a_1}} L_{f_{a_2}} \cdots L_{f_{a_L}} h\big)(x_0), \tag{3}$$

summed over all words [9, 19]; without a drift the same formula holds with the channel letters alone. Here $L_f g = \nabla g \cdot f$ is the rate of change of $g$ along $f$, the Lie derivative. The coefficient of a word measures how strongly that sequence of actions changes the output, starting from $x_0$. It is a single number for each word. The coefficients depend on the initial state $x_0$ but not on the input.

Many coefficients are zero at every state, because of the structure of the vector fields. In the forced Duffing oscillator below, for example, the input enters only the equation for $q_2$ and the output is $q_1$, so $L_{f_1} h = 0$; since the last letter of a word acts first on $h$, every word whose last letter is the input letter has a zero coefficient. Of the 30 words of length 1 to 4 over two letters, only 7 have non-zero coefficients for this system, and 16 of the 126 words up to length 6. A fit can represent this pattern only if it can set coefficients exactly to zero; Section 5.3 shows how far word selection gets.

The coefficients, and so the size of the smallest realisation of the series, belong to the starting state. At special states the size can be smaller: at an equilibrium, for instance, several Lie derivatives vanish together, and the rank of the Hankel matrix of the series (Section 3) drops. A recording that passes through different regimes therefore has a local rank that changes along it; the companion note on layered automata shows this for an oscillator driven through a bifurcation [15].

The empty word has $I_{()} = 1$, so its coefficient $c_{()} = h(x_0) = y(0)$ is a constant: the output at the start of the window. Each window of data is therefore shifted to start at zero before it is fitted, and the series is fitted to $y(t) - y(0)$, with $c_{()} = 0$. The model then predicts $y(t) - y(0)$, and adding the first sample of the window, $y(0)$, gives $y(t)$. This is done separately for every window,

because each window starts from its own state $x_0$ and so has its own $y(0)$. The realisations below are built for the shifted series in the same way. Placing $y(0)$ in the empty-word coefficient instead gives realisations of the same size for the driven Duffing series, at every order in Table 1.

The truncated series of order $K$ keeps the words up to length $K$ and sets every longer coefficient to zero. It approximates the output over a short enough window.

# 3 Bilinear realisations of a truncated series

## 3.1 Bilinear realisations

A bilinear system has a state that evolves linearly at each instant, with matrices that depend on the input:

$$\dot{z} = A_0 z + \sum_{i=1}^{m} u_i(t)\, A_i z, \qquad \hat{y} = \mu^\top z, \qquad z(0) = \lambda. \tag{4}$$

Its output also has a Chen–Fliess series, and its coefficients are products of its matrices:

$$c_w = \mu^\top A_{a_L} \cdots A_{a_1} \lambda \qquad \text{for } w = (a_1, \ldots, a_L). \tag{5}$$

Reading a word letter by letter moves the state, $z \mapsto A_a z$, and the read-out $\mu$ turns the final state into a number. A realisation of a set of coefficients is a choice of $(\lambda, \{A_a\}, \mu)$ that reproduces every coefficient through (5). A realisation is a finite model. It can be integrated as an ordinary differential equation for any new input, and its kernels follow from its matrices.

A realisation is reachable if the states $A_{a_L} \cdots A_{a_1} \lambda$, over all words, span the whole state space. It is observable if the read-outs $\mu^\top A_{a_L} \cdots A_{a_1}$ span the whole dual space. It is minimal if no realisation with fewer states reproduces the same coefficients. As in the linear case, a realisation is minimal if and only if it is reachable and observable, and two minimal realisations differ only by a change of basis [7, 1].

## 3.2 The Hankel matrix

The Hankel matrix of a set of coefficients has one row and one column for each word. The entry in row $u$ and column $v$ is the coefficient of the word $uv$, $u$ followed by $v$:

$$H[u, v] = c_{uv}. \tag{6}$$

For a realisation, $c_{uv} = (\mu^\top A_v)(A_u \lambda)$, where $A_u$ is the product along $u$. So $H$ is a product of a matrix with $n$ columns and a matrix with $n$ rows, and its rank is at most $n$. Fliess showed that the rank of the Hankel matrix is exactly the size of a minimal realisation [8]. A factorisation $H = PQ$ of rank $r$ gives one. With $H_a[u, v] = c_{uav}$, the matrices are $A_a = (P^+ H_a Q^+)^\top$, the initial state is the row of $P$ for the empty word, and the read-out is the column of $Q$ for the empty word.

For a series truncated at order $K$, the staircase Hankel matrix is defined as follows: the rows and columns are all words of length 0 to $K$, and

$$H[u, v] = \begin{cases} c_{uv}, & |u| + |v| \le K, \\ 0, & |u| + |v| > K. \end{cases} \tag{7}$$

The zeros are not a choice; they are the coefficients of the truncated series beyond order $K$. A row word of length $\ell$ has non-zero entries only in columns of length at most $K - \ell$, so the non-zero entries form a staircase. The same holds for every $H_a$. Every entry of $H$ and of every $H_a$ is therefore known: either a coefficient up to order $K$, or a zero.

*Example* 2 (The staircase Hankel matrix at $K = 2$). For the Duffing oscillator of Section 4, truncated at $K = 2$, the non-zero coefficients are $c_{(0)} = 0.3$, $c_{(0,0)} = 0.285$, $c_{(1,0)} = 1$. The words up to length 2 are (), (0), (1), $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$, and the staircase Hankel matrix is

| | () | (0) | (1) | (0,0) | (0,1) | (1,0) | (1,1) |
|---|---|---|---|---|---|---|---|
| () | 0 | 0.3 | 0 | 0.285 | 0 | 1 | 0 |
| (0) | 0.3 | 0.285 | 0 | 0 | 0 | 0 | 0 |
| (1) | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| (0,0) | 0.285 | 0 | 0 | 0 | 0 | 0 | 0 |
| (0,1) | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| (1,0) | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| (1,1) | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Its rank is 3, so the truncated series has a minimal realisation with 3 states.

### 3.3 Truncation and nilpotency

**Definition 1.** *A realisation is* nilpotent of index $K + 1$ *if* $A_{a_{K+1}} \cdots A_{a_1} = 0$ *for every choice of* $K + 1$ *letters. (Shorter products may also vanish; the index here is the length named, not necessarily the smallest one.)*

A nilpotent realisation reads any word longer than $K$ to zero. Its coefficients beyond order $K$ are therefore zero, as those of a truncated series are.

**Proposition 1.** *A minimal realisation of a series truncated at order* $K$ *is nilpotent of index* $K + 1$.

*Proof.* Let $w$ be a word of length $K + 1$. For all words $u$ and $v$, $\mu^\top A_v A_w A_u \lambda = c_{uwv} = 0$, because $|uwv| > K$. Since the realisation is reachable, the vectors $A_u\lambda$ span the state space, so $\mu^\top A_v A_w = 0$ for every $v$. Since it is observable, the vectors $A_v^\top \mu$ span the state space, so $A_w = 0$. □

*Remark* 1. A realisation that is not minimal can reproduce a truncated series without being nilpotent, if it contains states that the input cannot reach or the output cannot see. Its coefficients beyond order $K$ are still zero, so its output still contains no terms beyond order $K$. Nilpotency is therefore sufficient for a realisation to have no terms beyond order $K$, but not necessary; it is guaranteed for minimal realisations. For a reduced realisation, which reproduces the series only approximately, nilpotency is the simple way to guarantee that no terms beyond order $K$ appear.

*Remark* 2 (Validity). The solution of (4) can be written as a sum over words of products of the matrices times iterated integrals, obtained by substituting the equation into itself repeatedly (the Peano–Baker series); its coefficients are given by (5). In a nilpotent realisation every product of more than $K$ matrices is zero, so this sum stops at length $K$, and the output is exactly the truncated series $\sum_{|w|\le K} c_w I_w(t)$, for all times and all inputs. The integrals of drift words are powers of time, $I_{(0,\ldots,0)}(t) = t^L/L!$, and the others are integrals of the input weighted by powers of time. Such an output need not blow up; for example, it stays bounded if every coefficient involving the drift letter vanishes and the iterated integrals of the inputs stay bounded. But its drift words are polynomials in time, which cannot decay, so outside the window it is in general a poor description of the system. It is also true that every matrix of a nilpotent realisation, and every combination $A_0 + \sum_i u_i A_i$, has all its eigenvalues equal to zero. That is a consequence of nilpotency, not a test for it: matrices that each have only zero eigenvalues can have products that never vanish. Nor are computed eigenvalues a reliable sign of nilpotency: a perturbation of size $\varepsilon$ of a nilpotent $n \times n$ matrix can move its eigenvalues by as much as $\varepsilon^{1/n}$, so the products themselves must be checked. A nilpotent realisation describes the system only over the window on which the truncated series does; Section 4.4 shows how this window ends in one example.

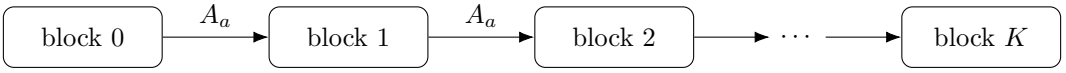



Figure 1: The graded state. Every letter moves the state one block forward. Nothing moves back, sideways or past block $K$.

### 3.4 The staircase realisation

The staircase Hankel matrix is factorised by a singular value decomposition, $H = U\Sigma V^\top$. This writes $H$ as a sum of simple rank-one pieces, ordered by their sizes $\sigma_1 \geq \sigma_2 \geq \ldots$, the singular values. The rank $r$ is the number of singular values above $\tau\sigma_1$, for a relative tolerance $\tau$; discarding the small pieces gives a smaller model of a nearby series. With $P = U_r\Sigma_r^{1/2}$ and $Q = \Sigma_r^{1/2}V_r^\top$, the realisation follows from the formulas of the previous section. When no non-zero singular value is discarded, this is a minimal realisation, and by Proposition 1 it is nilpotent. When singular values are discarded to obtain a smaller model, the result is a realisation of a nearby series of lower rank. That series is in general not truncated, and the realisation is no longer nilpotent.

### 3.5 The graded construction

The graded construction keeps the structure of the word-length grading at every step. Its state is divided into blocks, one for each word length $\ell = 0, \ldots, K$. Every matrix $A_a$ moves the state from block $\ell$ to block $\ell + 1$ only (Figure 1). After $K + 1$ letters there is no block left, and the state is zero.

The blocks are built from the longest words down. For a word $u$ of length $\ell$, its local future vector lists its own coefficient and the coordinates of its one-letter extensions in the already reduced block $\ell + 1$:

$$F_u = \big(c_u,\ \kappa_{\ell+1}(u0),\ \kappa_{\ell+1}(u1),\ \ldots,\ \kappa_{\ell+1}(um)\big), \tag{8}$$

where $\kappa_{\ell+1}(v)$ is the coordinate vector of the word $v$ in block $\ell + 1$ (empty when $\ell = K$). It is a compressed form of the row of $u$ in the staircase Hankel matrix: the row of $u$ is its own coefficient followed by the rows of its extensions $u0, \ldots, um$. For $\ell = K, K-1, \ldots, 0$:

1. stack the vectors $F_u$ of all words of length $\ell$ into a matrix $F_\ell$;
2. factorise $F_\ell = C_\ell R_\ell$ by a singular value decomposition, keeping the $r_\ell$ singular values above $\tau$ times the largest; the row of $C_\ell$ for the word $u$ is its coordinate vector $\kappa_\ell(u)$;
3. read the block-$\ell$ part of the read-out from the first column of $R_\ell$, and the block from $\ell$ to $\ell + 1$ of each $A_a$ from the columns of $R_\ell$ that belong to letter $a$, transposed.

The initial state is $\kappa_0(())$ in block 0. The total number of states is $n = r_0 + \cdots + r_K$.

*Example* 3 (The graded construction at $K = 2$). For the coefficients of Example 2, the four words of length 2 have local future vectors equal to their coefficients, $F_2 = (0.285,\ 0,\ 1,\ 0)^\top$, of rank 1. The two words of length 1 give $F_1 = \begin{pmatrix} 0.3 & 0.2795 & 0 \\ 0 & 0.9807 & 0 \end{pmatrix}$, of rank 2. The empty word gives a single row $F_0$, of rank 1. The realisation has $1 + 2 + 1 = 4$ states, one more than the minimal realisation of Example 2. Its matrices are given in Section 4.3.

### 3.6 Properties

**Proposition 2** (Nilpotency)**.** *For every tolerance $\tau$, the graded realisation is nilpotent of index $K + 1$.*

*Proof.* Every $A_a$ maps block $\ell$ into block $\ell+1$ and block $K$ to zero. A product of $K+1$ letters therefore maps every block to zero. The argument uses only this block structure, not the values in the blocks. □

Let $H_\ell$ denote the rows of the staircase Hankel matrix that belong to words of length $\ell$.

**Proposition 3** (Exactness and size)**.** *If no non-zero singular value is discarded ($\tau \to 0$), the graded realisation reproduces every coefficient up to order $K$ exactly, and $r_\ell = \operatorname{rank} H_\ell$ for every $\ell$. Its size is therefore $n = \sum_{\ell=0}^{K} \operatorname{rank} H_\ell$, which is at least $\operatorname{rank} H$, the minimal size.*

*Proof.* The row of a word $u$ of length $\ell$ in the staircase matrix is $h_u = (c_u, h_{u0}, \ldots, h_{um})$, because $c_{(ua)v}$ is the entry of the row of $ua$ in column $v$, and every column beyond the staircase is zero. We show by induction from $\ell = K$ down that $h_u = \kappa_\ell(u) M_\ell$ for a matrix $M_\ell$ of full row rank $r_\ell$. For $\ell = K$, $h_u = (c_u) = F_u = \kappa_K(u) R_K$, so $M_K = R_K$. At length $\ell$, $h_u = F_u\, D_{\ell+1}$, where $D_{\ell+1}$ is block diagonal with blocks $1, M_{\ell+1}, \ldots, M_{\ell+1}$, and $F_u = \kappa_\ell(u) R_\ell$ when nothing is discarded. So $M_\ell = R_\ell D_{\ell+1}$, which has full row rank because $R_\ell$ and $D_{\ell+1}$ do. Hence $\operatorname{rank} H_\ell = \operatorname{rank} F_\ell = r_\ell$. For exactness, place $\kappa_\ell(u)$ in block $\ell$; this is the state after reading $u$. Reading a further letter $a$ gives the block of $R_\ell$ for $a$ applied to $\kappa_\ell(u)$, which is the part of $F_u$ for $a$, that is $\kappa_{\ell+1}(ua)$. The read-out of block $\ell$ gives the first entry of $F_u$, which is $c_u$. □

When nothing is discarded, both constructions are nilpotent and both reproduce the series exactly. They differ in size by $\sum_\ell \operatorname{rank} H_\ell - \operatorname{rank} H$. The difference is the dependence between rows of different word lengths, which the graded construction cannot use, because it never mixes blocks. They differ in what happens when singular values are discarded to obtain a smaller model. The staircase realisation then stops being nilpotent, and its solution acquires terms beyond order $K$. The graded realisation stays nilpotent whatever is discarded.

# 4 Worked examples with exact coefficients

## 4.1 A synthetic system

A bilinear system with 4 states and strictly triangular random matrices is nilpotent, with a series that ends at order 3. From its coefficients alone, the staircase realisation recovers 4 states and the graded construction 6 (blocks (1,2,2,1)), against 15 for a realisation with one state per word. Both reproduce every coefficient to $6.7 \times 10^{-16}$. The graded size equals the sum of the block ranks of the staircase matrix, 6, as Proposition 3 requires.

## 4.2 The forced Duffing oscillator

The test system is

$$\dot{q}_1 = q_2, \qquad \dot{q}_2 = -\delta q_2 - \alpha q_1 - \beta q_1^3 + u(t), \qquad y = q_1, \tag{9}$$

with $\delta = 0.3$, $\alpha = -1$, $\beta = 1$ and $x_0 = (0.5, 0.3)$. With $\alpha < 0$ it has two wells; it has no closed-form solution. The coefficients were computed exactly from (3) by symbolic Lie derivatives, with the parameters as exact fractions, and then rounded to double precision. Every reference solution was computed by the eighth-order Dormand–Prince method with a relative tolerance of $10^{-13}$, and the iterated integrals by the exact method of Section 2.2.

Table 1 gives the sizes for $K = 2$ to 8. The staircase rank was computed by a singular value decomposition and certified by exact Gaussian elimination modulo a large prime, applied to the rounded coefficients; the two agree at every order. At $K = 4$ and 6 a minimal realisation was also built in exact rational arithmetic, and at $K = 8$ in floating point, and its products of $K+1$ matrices were checked to vanish. The graded size equals $\sum_\ell \operatorname{rank} H_\ell$ at every order, as Proposition 3 states, and is 1.33-1.89 times the minimal size. Both constructions reproduce the coefficients to rounding error.

Table 1: Sizes of the realisations of the Duffing series truncated at order $K$. Word tree: one state per word. Graded: block sizes $r_0, \ldots, r_K$ and total. Error: largest difference between a reproduced and a true coefficient. Nilpotency: largest norm of a product of $K+1$ matrices.

| $K$ | word tree | staircase rank | minimal | graded blocks | graded | error | nilpotency (staircase) |
|---|---|---|---|---|---|---|---|
| 2 | 7 | 3 | | (1,2,1) | 4 | $4.4 \times 10^{-16}$ | $5.3 \times 10^{-16}$ |
| 3 | 15 | 4 | | (1,2,2,1) | 6 | $6.4 \times 10^{-16}$ | $1.5 \times 10^{-16}$ |
| 4 | 31 | 5 | 5 | (1,2,2,2,1) | 8 | $8.3 \times 10^{-16}$ | $5.4 \times 10^{-16}$ |
| 5 | 63 | 7 | | (1,2,3,3,2,1) | 12 | $4.0 \times 10^{-15}$ | $1.6 \times 10^{-15}$ |
| 6 | 127 | 9 | 9 | (1,2,4,4,3,2,1) | 17 | $3.4 \times 10^{-14}$ | $2.3 \times 10^{-15}$ |
| 7 | 255 | 12 | | (1,2,4,5,4,3,2,1) | 22 | $3.6 \times 10^{-14}$ | $3.1 \times 10^{-15}$ |
| 8 | 511 | 16 | 16 | (1,2,4,7,6,4,3,2,1) | 30 | $1.2 \times 10^{-13}$ | $1.7 \times 10^{-15}$ |

Table 2: Largest absolute differences over 0.15 s, exact coefficients.

| $K$ | input | graded − true | staircase − true | graded − series | staircase − series |
|---|---|---|---|---|---|
| 6 | first | $1.4 \times 10^{-9}$ | $1.4 \times 10^{-9}$ | $1.2 \times 10^{-14}$ | $3.0 \times 10^{-14}$ |
| 6 | second | $4.9 \times 10^{-10}$ | $4.9 \times 10^{-10}$ | $1.1 \times 10^{-14}$ | $3.9 \times 10^{-14}$ |
| 8 | first | $1.1 \times 10^{-11}$ | $1.1 \times 10^{-11}$ | $1.4 \times 10^{-14}$ | $2.0 \times 10^{-14}$ |
| 8 | second | $6.3 \times 10^{-12}$ | $6.3 \times 10^{-12}$ | $1.2 \times 10^{-14}$ | $2.0 \times 10^{-14}$ |

### 4.3 The realised system at $K = 2$

The graded realisation of Example 3 is the bilinear system (4) with

$$A_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.2814 & 0 & 0 & 0 \\ 0.4784 & 0 & 0 & 0 \\ 0 & -1.0077 & 0.0468 & 0 \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.9026 & 0 & 0 & 0 \\ -0.1492 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\lambda = \begin{pmatrix} -1.0700 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mu = \begin{pmatrix} 0 \\ -0.0883 \\ -0.5341 \\ 1.0197 \end{pmatrix}$$

Integrated with the input $u(t) = 0.3 \sin 6\pi t + 0.1 \sin 14\pi t$ over 0.15 s, it differs from the true response by at most $4.5 \times 10^{-5}$, against a largest response of 0.051. This is the error of truncation at order 2.

### 4.4 Solutions over the window and beyond

At $K = 6$ and 8 the realisations were integrated over 0.15 s with the input above and with a second input, $u(t) = 0.25 \sin(9\pi t + 0.7) - 0.15 \cos 4\pi t$. Table 2 compares them with the true response and with the truncated series itself. Graded and staircase realisations agree with the series to about $10^{-14}$, so neither adds terms beyond order $K$. Their error against the true response is that of the truncated series: $1.4 \times 10^{-9}$ at $K = 6$ and $1.1 \times 10^{-11}$ at $K = 8$, for a response of about 0.05.

Table 3 follows the $K = 6$ graded realisation over longer times and larger inputs. Accuracy is lost as time grows, much more than as the input grows. Over 10 s the true output stays below 1.27 in magnitude, while the realised output reaches 2799. Here the realised output has moved far from the true response. This is typical, but not necessary.

Table 3: The $K = 6$ graded realisation over longer horizons and larger inputs, $u(t) = A(\sin 6\pi t + 0.3 \sin 14\pi t)$. Entries are $R^2$ against the true response.

| horizon | $A = 0.3$ | $A = 1$ | $A = 3$ |
|---|---|---|---|
| 0.15 s | 1.0000 | 1.0000 | 1.0000 |
| 0.3 s | 1.0000 | 1.0000 | 1.0000 |
| 0.5 s | 1.0000 | 1.0000 | 1.0000 |
| 1 s | 1.0000 | 1.0000 | 1.0000 |
| 2 s | 0.9903 | 0.9863 | 0.9637 |
| 3 s | −5.5123 | −8.1418 | −20.2158 |

### 4.5 A system without a letter for time

The driftless Duffing system

$$\dot{q}_1 = u_1, \qquad \dot{q}_2 = u_2 + \beta q_1^3 u_1 - \delta q_2 u_1, \qquad y = q_2, \tag{10}$$

with $\delta = 0.3$, $\beta = 1$ and $x_0 = (0.5, 0.3)$, has the cubic term and damping of the driven oscillator but no drift. Its alphabet has two letters, one for each channel, written 1 and 2. Its series does not end at any finite length: symbolically, exactly two words of each length from 1 to 10 have non-zero coefficients. Table 4 gives the sizes of its realisations. The staircase ranks grow by one with each order, from 3 at $K = 2$ to 9 at $K = 8$, certified as before, and the graded sizes from 4 to 12.

At $K = 2$ the non-zero coefficients are $c_{(1)} = 0.035$, $c_{(2)} = 1$, $c_{(1,1)} = 0.7395$, $c_{(2,1)} = -0.3$, and the graded construction gives four states in blocks $(1, 2, 1)$ with

$$A_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.3713 & 0 & 0 & 0 \\ -0.6792 & 0 & 0 & 0 \\ 0 & -0.6157 & 0.7063 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -0.8197 & 0 & 0 & 0 \\ -0.3077 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\lambda = \begin{pmatrix} -1.1686 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mu = \begin{pmatrix} 0 \\ 0.8525 \\ 0.5102 \\ 0.8933 \end{pmatrix}$$

The state of block 0 receives nothing, so it stays at its initial value, and the realisation reduces to three equations,

$$\begin{aligned} \dot{z}_1 &= -0.4339\, u_1(t) + 0.9579\, u_2(t) \\ \dot{z}_2 &= 0.7937\, u_1(t) + 0.3596\, u_2(t) \\ \dot{z}_3 &= -0.6157\, z_1\, u_1(t) + 0.7063\, z_2\, u_1(t) \\ y(t) - y(0) &= 0.8525\, z_1 + 0.5102\, z_2 + 0.8933\, z_3 \end{aligned}$$

The first two states integrate the inputs; the third multiplies them by the first channel. With no input, nothing moves.

Choosing the rank needs care even with exact coefficients. At $K = 6$ the singular values of the staircase Hankel matrix are 7.27, 6.51, 5.82, 5.35, 5.14, 0.00267, 0.000623, followed by values below $10^{-15}$. A tolerance of $\tau = 10^{-2}$ keeps 5 and misses the two small but genuine singular values; $\tau = 10^{-6}$ keeps the correct 7.

Without a drift, time enters only through the path $\int_0^t u$. The error of the truncated series depends on the size of the iterated integrals of the path: how far it moves, and also the areas it sweeps, which keep growing when the path loops even if it stays near its start. Table 5 shows the $K = 6$ realisations driven by $u = (0.3A \sin 6\pi t + b,\ 0.1A \sin 14\pi t)$, and by inputs that move the

Table 4: Sizes of the realisations of the driftless Duffing series truncated at order $K$, with the same columns as Table 1.

| $K$ | word tree | staircase rank | graded blocks | graded | error | nilpotency (staircase) |
|---|---|---|---|---|---|---|
| 2 | 7 | 3 | (1,2,1) | 4 | $3.8 \times 10^{-16}$ | $3.7 \times 10^{-16}$ |
| 3 | 15 | 4 | (1,2,2,1) | 6 | $2.3 \times 10^{-15}$ | $1.1 \times 10^{-15}$ |
| 4 | 31 | 5 | (1,2,2,2,1) | 8 | $9.8 \times 10^{-15}$ | $2.0 \times 10^{-15}$ |
| 5 | 63 | 6 | (1,2,2,2,1,1) | 9 | $2.8 \times 10^{-14}$ | $2.9 \times 10^{-15}$ |
| 6 | 127 | 7 | (1,2,2,2,1,1,1) | 10 | $2.2 \times 10^{-14}$ | $1.3 \times 10^{-15}$ |
| 7 | 255 | 8 | (1,2,2,2,1,1,1,1) | 11 | $1.6 \times 10^{-14}$ | $1.0 \times 10^{-14}$ |
| 8 | 511 | 9 | (1,2,2,2,1,1,1,1,1) | 12 | $1.3 \times 10^{-14}$ | $1.2 \times 10^{-15}$ |

Table 5: The driftless Duffing series truncated at $K = 6$ and its realisations. Excursion: largest value of $|\int_0^t u_i|$ over both channels. Area: largest absolute signed area swept by the path, $\frac{1}{2}|I_{(1,2)} - I_{(2,1)}|$. Errors are largest absolute differences.

| input | duration | excursion | area | largest output | series − true | graded − series | staircase − series |
|---|---|---|---|---|---|---|---|
| sines, $A = 1$ | 10 s | 0.032 | $8.4 \times 10^{-5}$ | 0.00603 | $1.6 \times 10^{-14}$ | $1.6 \times 10^{-14}$ | $8.6 \times 10^{-13}$ |
| sines, $A = 10$ | 10 s | 0.318 | $8.4 \times 10^{-3}$ | 0.109 | $9.0 \times 10^{-9}$ | $2.9 \times 10^{-13}$ | $3.1 \times 10^{-12}$ |
| sines, $A = 30$ | 10 s | 0.955 | $7.6 \times 10^{-2}$ | 1.06 | $1.9 \times 10^{-5}$ | $8.5 \times 10^{-12}$ | $9.4 \times 10^{-12}$ |
| sines, $A = 100$ | 10 s | 3.183 | $8.4 \times 10^{-1}$ | 37.6 | $8.2 \times 10^{-2}$ | $9.3 \times 10^{-10}$ | $9.3 \times 10^{-10}$ |
| sines, $A = 1$, $b = 0.1$ | 10 s | 1.016 | $2.3 \times 10^{-3}$ | 1.12 | $3.0 \times 10^{-5}$ | $5.4 \times 10^{-14}$ | $1.1 \times 10^{-12}$ |
| sines, $A = 1$, $b = 0.3$ | 10 s | 3.000 | $6.8 \times 10^{-3}$ | 30.6 | $5.4 \times 10^{-2}$ | $3.6 \times 10^{-13}$ | $1.5 \times 10^{-12}$ |
| circle | 10 s | 0.400 | $1.3 \times 10^{0}$ | 0.559 | $5.6 \times 10^{-8}$ | $3.7 \times 10^{-12}$ | $3.7 \times 10^{-12}$ |
| circle | 100 s | 0.400 | $1.3 \times 10^{1}$ | 4.07 | $1.5 \times 10^{-7}$ | $2.5 \times 10^{-10}$ | $2.3 \times 10^{-10}$ |

path round a circle of radius 0.2 once a second. With zero-mean sinusoidal inputs of moderate size the path stays near its start, sweeps little area, and the truncated series stays accurate for the whole 10 s. Larger inputs, or a constant offset $b$ that carries the path away, bring in the truncation error. On the circle the path never moves more than 0.4 from its start, yet the area it sweeps grows tenfold from 10 s to 100 s, and the output and the error of the truncated series grow with it, though more slowly. In every case both realisations agree with the truncated series to rounding error.

### 4.6 A finite series: the Heisenberg integrator

The Heisenberg, or Brockett, integrator

$$\dot{q}_1 = u_1, \qquad \dot{q}_2 = u_2, \qquad \dot{q}_3 = q_1 u_2 - q_2 u_1, \qquad y = q_3,$$

from $x_0 = (0.3, 0.5, 0)$, is the standard example of a driftless system with a nilpotent Lie algebra. Its series ends: every coefficient of a word of length 3 or more is zero (checked symbolically up to length 6). This is because the vector fields are affine in the state and the output is a state: two Lie derivatives of $q_3$ leave a constant, and a third leaves zero. A nilpotent Lie algebra alone would not be enough with a more complicated output. Truncation at $K = 2$ is therefore not an approximation, and the realisation of the truncated series reproduces the system for all times. Its staircase rank is 4 and its graded size 4. Driven by sinusoidal inputs of size 5 for 50 s, and by inputs that move the path round a circle for 100 s, sweeping an area that grows without limit, both realisations stay within $1.4 \times 10^{-10}$ of the true output, which reaches 25.2. This is the one case in this tutorial where a nilpotent realisation is valid beyond any window.

Table 6: Sizes of the realisations of the four-channel series truncated at order $K$.

| $K$ | word tree | staircase rank | graded blocks | graded |
|---|---|---|---|---|
| 2 | 21 | 6 | (1,4,1) | 6 |
| 3 | 85 | 10 | (1,4,5,1) | 11 |
| 4 | 341 | 24 | (1,4,14,5,1) | 25 |
| 5 | 1365 | 38 | (1,4,14,19,5,1) | 44 |
| 6 | 5461 | 69 | (1,4,14,32,19,5,1) | 76 |

### 4.7 Four input channels

To show how the sizes grow with the alphabet, Table 6 gives them for a system with four channels and trigonometric couplings,

$$\dot{q}_1 = u_1 + \sin(q_3)u_2, \quad \dot{q}_2 = u_2 + \cos(q_4)u_3, \quad \dot{q}_3 = u_3 + \sin(q_1)u_4, \quad \dot{q}_4 = u_4 + \cos(q_2)u_1,$$
$$y = q_1 q_2 + \sin q_3 + q_4^2,$$

from $x_0 = (0.3, 0.5, -0.2, 0.4)$. Its series does not end: 2913 of the 4096 words of length 6 have non-zero coefficients. A realisation with one state per word would need 5461 states at $K = 6$; the staircase realisation needs 69 and the graded one 76. At $K = 7$ the graded construction needs 126 states, against 21845 words, and takes well under a second; there the staircase matrix, with 21845 rows and columns, was not factorised.

## 5 Fitting a truncated series to data

### 5.1 Design

The coefficients of the series depend on the starting state but not on the input. Several trials that start from the same state, each with its own input, therefore share one set of coefficients. For the driven Duffing oscillator (one input and a letter for time) and the driftless Duffing system (two channels), we generated 24 trials from $x_0$, each of 0.15 s, sampled every 1 ms. Each channel of each trial was driven by its own sum of three sinusoids with random frequencies between 1 and 15 Hz, random phases and a standard deviation of 0.3. The same inputs were used at $K = 4$ and $K = 6$; the further studies below draw their own inputs. White noise, with a standard deviation given as a percentage of that of the response $y(t) - y(0)$, was added to every sample. Each noisy window was then shifted to start at zero by subtracting its own first sample, as is done with measured data. That sample is itself noisy, so the shift leaves an error that is constant over the window, and one free offset for each window, in effect an empty-word coefficient of its own, was fitted with the shared coefficients. The iterated integrals were computed by iisignature from the samples (Section 2.2). The trials were split into 12 training trials, 4 validation trials and 8 test trials.

The coefficients up to order $K$ were fitted in two steps, both using all training trials together. The first step chooses which words to keep. It starts with no words. At each stage it adds the word whose column of integrals, scaled to unit length, is most closely aligned with the part of the response that the words already chosen leave unexplained. It then refits all the chosen words by least squares. This form of forward selection is known as orthogonal matching pursuit. Each stage gives a candidate model with one more word. The second step chooses among these candidates on the validation trials, by the one-standard-error rule. The error of a candidate on a validation trial is its mean squared error after the trial's own mean is removed, since that mean is the free offset. The rule takes the candidate with the fewest words whose mean error over the validation trials is within one standard error of the smallest mean error. It prefers the simpler model when the data cannot tell the two apart. The chosen words are fitted by least squares

Table 7: $R^2$ on the test trials of the fitted series, with word selection (median over 20 noise draws; one fit without noise). Noise: standard deviation as a percentage of that of the response.

| system | $K$ | no noise | 0.1 % | 1 % | 3 % | 10 % |
|---|---|---|---|---|---|---|
| driven | 4 | 1.00000 | 1.00000 | 1.00000 | 0.99981 | 0.99857 |
| driven | 6 | 1.00000 | 1.00000 | 0.99998 | 0.99985 | 0.99841 |
| driftless | 4 | 1.00000 | 1.00000 | 0.99999 | 0.99971 | 0.99743 |
| driftless | 6 | 1.00000 | 1.00000 | 0.99999 | 0.99971 | 0.99721 |

with no penalty, so the coefficients that are kept are not shrunk, and every other coefficient is exactly zero. All results are on the test trials, which were used for nothing else. Accuracy is the $R^2$ on the test trials: one minus the squared error of the prediction divided by the variance of the true response, computed for each test trial and averaged; 1 is perfect. The true response here is the response without noise, so $R^2$ measures how well the fit recovers the system, and can be close to 1 even at high noise.

Ridge regression, which adds a cost on the sum of the squared coefficients, is a common alternative. It is not used here, for two reasons. Its cost never sets a coefficient exactly to zero, so it cannot represent the coefficients that are zero because of the structure of the system (Section 2.3). Selection can represent them, although Section 5.3 shows that it does not find them all. Ridge regression also shrinks the fitted coefficients as a whole towards zero, accepting a bias in return for less variance. Selection does not shrink the coefficients it keeps, but it has a bias of its own: the coefficients it keeps take up the effect of the words it leaves out (Section 5.5).

The companion note on layered automata [15] selects words in a different way, by sequential thresholding: it fits all words, drops those whose contribution to the response is small, refits, and repeats. That needs a fit of all words to start from. Here, at $K = 6$, with 126 coefficients and 12 short trials, the fit of all words is poorly determined (Section 5.5), and forward selection, which never fits all the words at once, is the safer choice. Both methods end with an unpenalised refit of the chosen words. Where the data contain occasional large errors, the squared error of the refit can be replaced by the Huber loss, which is squared for small residuals and grows only in proportion to the residual for large ones. The code provides it; the noise here is Gaussian, so it was not used.

### 5.2 How well the fitted series predicts

Table 7 gives the $R^2$ on the test trials. Without noise both systems are fitted to $R^2 = 1.00000$ at both orders. With noise the $R^2$ stays above 0.997 at $K = 6$ for both systems up to 10 % noise. At $K = 6$ and 10 % noise it is 0.9984 for the driven oscillator and 0.9972 for the driftless system.

Three choices in the design matter. Without the free offsets, the unexplained fraction of the variance of the response on the test trials, $1 - R^2$ (median at 1 % noise), grows from $1.9 \times 10^{-6}$ to $5.7 \times 10^{-5}$ at $K = 4$ and from $2.0 \times 10^{-5}$ to $4.9 \times 10^{-5}$ at $K = 6$ for the driven oscillator, and from $1.0 \times 10^{-5}$ to $1.5 \times 10^{-4}$ at $K = 6$ for the driftless system. Without word selection, ordinary least squares on all 126 words at $K = 6$ and 1 % noise gives a median $R^2$ of $-825$ (driven) and $-997.9$ (driftless) on the test trials, although the test inputs are of the same kind as the training inputs. A negative $R^2$ means the prediction is worse than predicting the mean of the response: the fit of all words follows the noise in the training trials and fails on new ones. The number of trials matters too. In a separate set of trials of the same design, at $K = 8$, 12 trials of 0.15 s are too few even to fit all the words. Each trial is a smooth function of time and adds only a limited number of independent directions, and the design, with 510 columns, has numerical rank 299. With 48 training trials it has full rank, 510.

Table 8: Words kept by the selection at $K = 6$: number kept (median, with the range over noise draws), how many of the exact non-zero words are missing, and how many words with exact coefficient zero are included (medians over 20 noise draws; one fit without noise).

| system | noise | words kept | exact words missing | extra words |
|---|---|---|---|---|
| driven (16 exact) | none | 58 | 5 | 47 |
| | 0.1 % | 30 (8-44) | 8 | 21.5 |
| | 1 % | 8 (5-11) | 12 | 4 |
| | 3 % | 5 (3-17) | 12.5 | 1.5 |
| | 10 % | 2 (2-5) | 14 | 0 |
| driftless (12 exact) | none | 21 | 5 | 14 |
| | 0.1 % | 8 | 7 | 3 |
| | 1 % | 7 (3-10) | 7 | 2.5 |
| | 3 % | 3 (2-3) | 9 | 0 |
| | 10 % | 2 (1-2) | 10 | 0 |

### 5.3 Which words are kept

Table 8 compares the words kept with the words whose exact coefficients are not zero. In none of the 324 fits of both systems at both orders did the two sets agree exactly. Without noise the fit keeps many more words than the exact series has: 58 words for the driven oscillator at $K = 6$, of which 47 have exact coefficients of zero. These extra words take up the error of truncation and of sampling the integrals, which are the only errors left. With noise the fit keeps fewer words: at $K = 6$ and 1 % noise it keeps a median of 8 of the 126 words for the driven oscillator, which has 16 non-zero coefficients, and 7 for the driftless system, which has 12.

The words left out are not always the ones with small effects. For the driven oscillator at $K = 6$ and 1 % noise, the word $(0, 0)$ is left out in 20 of 20 noise draws, although its exact term changes the response by 6.7 times the standard deviation of the noise (root-mean-square over the training trials). Its effect is taken up by other words made only of the time letter, whose integrals have nearly the same shape (Section 5.5). The fit then predicts well, but with the wrong coefficients for those words. The set of words kept is therefore a property of the data, its noise and its length, and of how well the columns can be told apart, as well as of the system.

### 5.4 The fitted coefficients

The coefficients multiply integrals of very different sizes, so they can only be compared on a common scale. The integral of a word of length $L$ with $n$ input letters has a size of order $A^n T^L / L!$, where $T = 0.15\,\mathrm{s}$ is the window and $A = 0.3$ the standard deviation of each input; this combines the two scales of Section 2. Each coefficient is multiplied by this size, and errors are given relative to the largest exact coefficient on this scale. Every letter of the driftless system is an input letter.

With selection, a coefficient can be wrong in two ways: a kept coefficient can have the wrong value, or a word with a non-zero exact coefficient can be left out, so that its error is its whole exact value. Table 9 gives the two separately, by word length, at $K = 6$ and 1 % noise. For the driven oscillator the median of the largest error of a kept coefficient is $6.8 \times 10^{-2}$ of the largest exact coefficient on this scale, for words of length 3; the largest coefficient left out is $7.1 \times 10^{-2}$, for words of length 2. The large errors are in words made only of the time letter: the largest error of a kept coefficient is $6.8 \times 10^{-2}$ for these words and $1.4 \times 10^{-3}$ for all other words (medians). The fitted series nevertheless predicts well (Table 7). The data fix the response, not the individual coefficients. The alphabet and the order fix which words the model may contain; the data choose among them and set the coefficients, and constrain them only through the response they produce.

Table 9: Errors of the fitted coefficients at $K = 6$ and 1 % noise, on the scale $c_w A^n T^L/L!$, relative to the largest exact coefficient on that scale (medians over 20 noise draws). Kept: largest error of a coefficient the fit kept; a dash means that in most draws no word of that length was kept. Left out: largest exact coefficient that the fit set to zero; a dash means that in most draws no non-zero word of that length was left out.

| | driven | | driftless | |
|---|---|---|---|---|
| length | kept | left out | kept | left out |
| 1 | $2.1 \times 10^{-2}$ | – | $7.1 \times 10^{-4}$ | – |
| 2 | $7.6 \times 10^{-4}$ | $7.1 \times 10^{-2}$ | $1.7 \times 10^{-3}$ | – |
| 3 | $6.8 \times 10^{-2}$ | $1.1 \times 10^{-3}$ | – | $9.4 \times 10^{-4}$ |
| 4 | $5.4 \times 10^{-4}$ | $9.2 \times 10^{-5}$ | $2.3 \times 10^{-3}$ | $1.0 \times 10^{-7}$ |
| 5 | – | $1.2 \times 10^{-5}$ | – | $5.3 \times 10^{-8}$ |
| 6 | $1.8 \times 10^{-2}$ | $2.8 \times 10^{-7}$ | – | $1.2 \times 10^{-10}$ |

### 5.5 Why the coefficients are hard to estimate

Five effects make the coefficients hard to estimate, and they act together. The numbers below are for the driven oscillator at $K = 6$ with all 126 words fitted by least squares, which shows the problem before any selection, and they are on the scale of Section 5.4, relative to the largest exact coefficient on that scale.

First, the terms of long words change the response very little. On this scale the largest exact coefficient of length 6 is $4.9 \times 10^{-7}$, so its term is far smaller than noise of 1 % of the response. At 1 % noise the predicted standard error of a coefficient is $4.5 \times 10^{-2}$ for words of length 1 and $3.6 \times 10^{1}$ for words of length 6.

Second, the columns of the design are nearly dependent. Each trial is a smooth function of time over a short window, so it resolves only a few independent shapes, while the number of words doubles with each extra letter. The condition number of the design, with its columns scaled to unit length, is $1.8 \times 10^{6}$ at $K = 6$, and at $K = 8$ the design does not have full rank (Section 5.2). Scaling the columns removes the size of the inputs, so larger inputs do not change this number: it is $1.8 \times 10^{6}$ with inputs ten times larger.

Third, the words made only of the time letter have integrals $t^L/L!$ that are the same in every trial, so adding trials does not help to tell them apart. After each window's mean is removed, which the free offset requires, they are polynomials in time that are nearly dependent on a short window: for lengths 1 to 6 the condition number of these six columns alone is $1.0 \times 10^{4}$. This is consistent with the largest errors in Table 9 being in these words.

Fourth, the series of the system does not end at $K$, and the fitted coefficients take up the error of truncation. With integrals computed exactly and no noise, the largest error of the coefficients of length 6 is $4.1 \times 10^{-8}$, which is 0.08 of the largest exact coefficient of that length.

Fifth, the integrals computed from samples are least accurate for long words (Section 2.2). For the inputs here the relative error of an integral has a median of $1.7 \times 10^{-5}$ at length 2 and $4.7 \times 10^{-4}$ at length 6, and reaches $1.7 \times 10^{-2}$ at length 6 for integrals whose largest value is very small. With sampled integrals and no noise, the largest error of the coefficients of length 6 is $3.8 \times 10^{-3}$, about 8000 times the largest exact coefficient of that length: here the sampling error, not the truncation, dominates the error without noise.

Longer windows, larger inputs and more trials all help. At 1 % noise the predicted standard error of the coefficients of length 6, divided by the largest exact coefficient of that length, is $7.3 \times 10^{7}$. It falls to $1.5 \times 10^{4}$ with windows of 0.6 s, to $2.2 \times 10^{6}$ with inputs ten times larger, and to $2.7 \times 10^{5}$ with 48 training trials. It stays far larger than 1, so none of these is a cure. Word selection does not estimate these coefficients either. It leaves out words whose effect it cannot separate from the noise or from that of other words, and the coefficients it keeps then take up the effect of the words left out, so they are biased. The selected model predicts the

Table 10: Largest possible sizes for a series truncated at order $K$ with two letters ($c_{()} = 0$), found from random coefficients, and the sizes for the exact driven Duffing coefficients.

| | largest possible | | | exact Duffing | | |
|---|---|---|---|---|---|---|
| $K$ | staircase | graded | Lie–Hankel | staircase | graded | Lie–Hankel |
| 2 | 4 | 4 | 3 | 3 | 4 | 2 |
| 3 | 6 | 7 | 4 | 4 | 6 | 3 |
| 4 | 10 | 11 | 6 | 5 | 8 | 4 |
| 5 | 14 | 18 | 8 | 7 | 12 | 6 |
| 6 | 22 | 26 | 11 | 9 | 17 | 7 |
| 7 | 30 | 41 | 15 | 12 | 22 | 8 |
| 8 | 46 | 57 | 21 | 16 | 30 | 9 |

response; its coefficients are not those of the system.

### 5.6 Sizes of realisations of fitted series

For coefficients drawn at random, the staircase rank, the graded size and the Lie–Hankel rank of Section 7 take the largest values possible for a series truncated at order $K$ with two letters (Table 10). When all words are fitted by least squares, the staircase realisations of the fitted series reach this ceiling: 10 states at $K = 4$ and 22 at $K = 6$, in every one of the 324 fits of both systems, at every noise level including no noise. The errors in the fitted coefficients fill every direction the truncated structure allows, and the size says nothing about the system.

With word selection the sizes follow the number of words kept, and they do not measure the system either. Without noise, many extra words are kept and the sizes reach the ceiling: 10 and 22 states for the driven oscillator at $K = 4$ and $K = 6$. As the noise grows, fewer words are kept and the sizes fall. For the driven oscillator at $K = 6$, the median staircase size is 21 at 0.1 % noise, 14 at 1 % and 3 at 10 %, against 9 for the exact truncated series and a ceiling of 22. For the driftless system the sizes are 18 without noise, 8 at 1 % and 2 at 10 %, against 7 for the exact series. At high noise the words left out pull the size below the exact value. The size of a fitted realisation is therefore a property of the system, the window, the noise and the selection together. Before it is used as a measure, for example to compare recordings, it should be compared with the ceiling, and the recordings compared should have the same noise level, length and selection rule.

### 5.7 Graded and staircase realisations of fitted coefficients

For the driven oscillator, both constructions were applied to every fitted series of a second study of the same design, with its own random inputs and 50 noise draws, with the same tolerance $\tau$ for discarding small singular values, and integrated with each test input. Table 11 and Figure 2 give the results. With $\tau = 10^{-6}$ the two realisations give the same responses on the test trials as the fitted series, to within $7.8 \times 10^{-9}$ in $R^2$, at every noise level. For the graded realisation the difference is at most $6.7 \times 10^{-16}$; the larger value comes from staircase realisations and is rounding error in the factorisation.

At 1 % and 10 % noise the selected fits are small, and the larger tolerance $\tau = 10^{-2}$ leaves the median sizes at $K = 6$ unchanged. At 0.1 % noise, where more words are kept and the sizes are close to the ceiling, the median size falls from 19 to 18.5 for the staircase realisation and from 24 to 23 for the graded one. Within the window, the two reduced realisations are equally accurate: at $K = 6$ their median $R^2$ values differ by at most $1.2 \times 10^{-5}$ at any noise level.

The two differ in what their solutions contain. The graded realisation assigns exactly zero to every word of length $K + 1$ to $K + 4$. This holds for the exact series and for fits at 0.1 %, 1 % and 10 % noise (one draw each, $K = 4$ and 6, both tolerances). Its integrated output equals

Table 11: $R^2$ on the test trials (median over 50 noise draws, with the 5th and 95th percentiles) and number of states (median), window 0.15 s, for realisations of the fitted series with word selection. Noise: standard deviation as a percentage of that of the response. The largest possible sizes are 10 (staircase) and 11 (graded) at $K = 4$, and 22 and 26 at $K = 6$ (Table 10).

| $K$ | $\tau$ | noise | graded $R^2$ | graded states | staircase $R^2$ | staircase states |
|---|---|---|---|---|---|---|
| 4 | $10^{-6}$ | 0.1 % | 1.0000 (1.000-1.0000) | 10 | 1.0000 (1.000-1.0000) | 8 |
| 4 | $10^{-6}$ | 1 % | 1.0000 (1.000-1.0000) | 10 | 1.0000 (1.000-1.0000) | 8 |
| 4 | $10^{-6}$ | 10 % | 0.9983 (0.998-0.9999) | 4 | 0.9983 (0.998-0.9999) | 3 |
| 4 | $10^{-2}$ | 0.1 % | 1.0000 (1.000-1.0000) | 9 | 1.0000 (1.000-1.0000) | 8 |
| 4 | $10^{-2}$ | 1 % | 1.0000 (1.000-1.0000) | 9.5 | 1.0000 (1.000-1.0000) | 8 |
| 4 | $10^{-2}$ | 10 % | 0.9983 (0.998-0.9999) | 4 | 0.9983 (0.998-0.9999) | 3 |
| 6 | $10^{-6}$ | 0.1 % | 1.0000 (1.000-1.0000) | 24 | 1.0000 (1.000-1.0000) | 19 |
| 6 | $10^{-6}$ | 1 % | 1.0000 (1.000-1.0000) | 9 | 1.0000 (1.000-1.0000) | 7 |
| 6 | $10^{-6}$ | 10 % | 0.9984 (0.996-0.9997) | 4 | 0.9984 (0.996-0.9997) | 3 |
| 6 | $10^{-2}$ | 0.1 % | 1.0000 (1.000-1.0000) | 23 | 1.0000 (1.000-1.0000) | 18.5 |
| 6 | $10^{-2}$ | 1 % | 1.0000 (1.000-1.0000) | 9 | 1.0000 (1.000-1.0000) | 7 |
| 6 | $10^{-2}$ | 10 % | 0.9984 (0.996-0.9997) | 4 | 0.9984 (0.996-0.9997) | 3 |

its own truncated series to the accuracy of the integrator over 0.15 s, 1 s and 3 s. The staircase realisation does so to rounding error when nothing is discarded: its coefficients of length $K + 1$ to $K + 4$ are then at most $8.0 \times 10^{-15}$ times its largest coefficient. When it is reduced, it is no longer nilpotent: at $K = 6$, $\tau = 10^{-2}$ and 0.1 % noise, it implies coefficients for words of length 7, which a realisation of the series truncated at $K = 6$ should not have. Within the 0.15 s window its output departs from its own truncated series by only $8.4 \times 10^{-12}$, against a response of about 0.05. After 1 s the departure is $1.7 \times 10^{-6}$, still negligible against an output of 0.62. After 3 s it is $1.3 \times 10^{-3}$. By then the output of the truncated series itself has reached 22, while the true output $y(t) - y(0)$ stays below 0.8 in absolute value. The series is then no longer a useful description, and the departure does not matter. In these examples, then, the structural guarantee of the graded construction did not change any prediction that the series could make.

### 5.8 One starting state

The coefficients describe the system from one starting state only, because they are Lie derivatives evaluated at $x_0$. To show this, the coefficients of the driven oscillator fitted at $K = 6$ without noise to trials that start from $x_0$ were used, unchanged, to predict trials that start from $(0.6, 0.2)$. On those trials the $R^2$ is 0.266, against 1.0000 on test trials that start from $x_0$. The fit itself is accurate; a series fitted in one window does not apply to a window that starts from another state.

## 6 Volterra kernels from the coefficients

The series (3) contains the Volterra kernels. With one input, a word with $n$ input letters is fixed by the numbers of drift letters before, between and after them:

$$w = (0^{j_0}, 1, 0^{j_1}, 1, \ldots, 1, 0^{j_n}).$$

**Proposition 4** (Kernels from words)**.** *For such a word,*

$$I_w(t) = \int_{0<s_1<\cdots<s_n<t} K_w(t;s)\, u(s_1)\cdots u(s_n)\, ds,$$

$$K_w(t;s) = \frac{s_1^{j_0}}{j_0!}\,\frac{(s_2 - s_1)^{j_1}}{j_1!}\cdots\frac{(t - s_n)^{j_n}}{j_n!}.$$

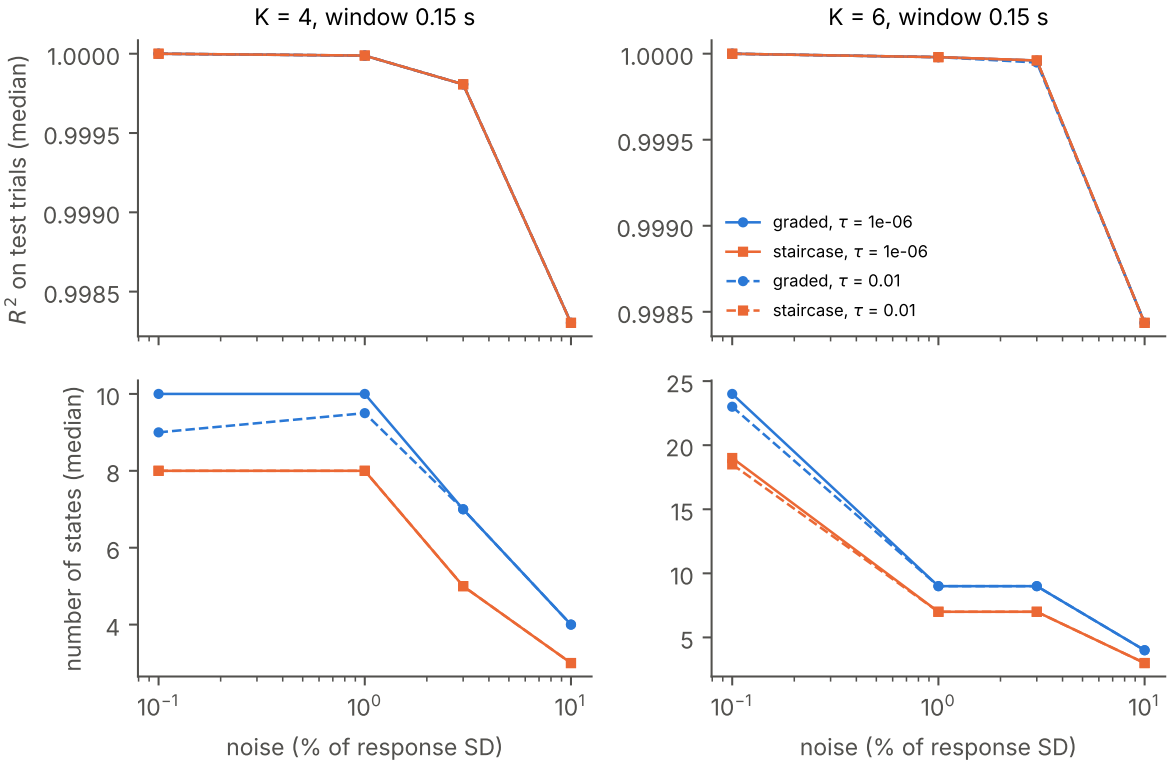



Figure 2: $R^2$ on the test trials and number of states against noise, for the graded and staircase realisations of fitted coefficients with word selection, at two tolerances (medians over 50 noise draws; window 0.15 s).

*Summing over the words with $n$ input letters, the $n$-th order Volterra kernel of the series is defined for all input times $s_1, \dots, s_n$ in $[0, t]$ by*

$$h_n(t; s_1, \dots, s_n) = \frac{1}{n!} \sum_w c_w K_w(t; s_{(1)}, \dots, s_{(n)}),$$

*where $s_{(1)} \le \cdots \le s_{(n)}$ are the same times in increasing order. It is symmetric in its arguments, and the order-$n$ part of the response is*

$$\int_0^t \cdots \int_0^t h_n(t; s_1, \dots, s_n)\, u(s_1) \cdots u(s_n)\, ds_1 \cdots ds_n.$$

*Proof.* Each drift letter integrates once against time, and the integrals of a run of $j$ drift letters between two input times give $(\Delta s)^j/j!$. Each input letter fixes one of the times $s_i$. The cube $[0, t]^n$ is the union of $n!$ ordered pieces, one for each order of the times, and the symmetric kernel takes the same values on each; the factor $1/n!$ makes the integral over the whole cube equal to the integral over one piece. □

The kernels depend on $x_0$, because the coefficients do.

At $K = 8$ the Duffing series has 8, 17, 16 and 5 non-zero words with 0, 1, 2 and 3 input letters, and none with 4. For the input $u(t) = 2 \sin 6\pi t + 0.5 \sin 14\pi t$ at $t = 0.1$ s, the contribution of each order computed from its kernel, by Gauss–Legendre quadrature on the ordered region, agrees with the contribution computed from the iterated integrals to $1.6 \times 10^{-17}$ or better. The contributions of orders 2 and 3 are $-1.6 \times 10^{-8}$ and $-3.7 \times 10^{-11}$.

The kernels are checked against the system itself, and only as a whole. A single kernel is not a response that the oscillator produces on its own, so the system offers nothing to compare

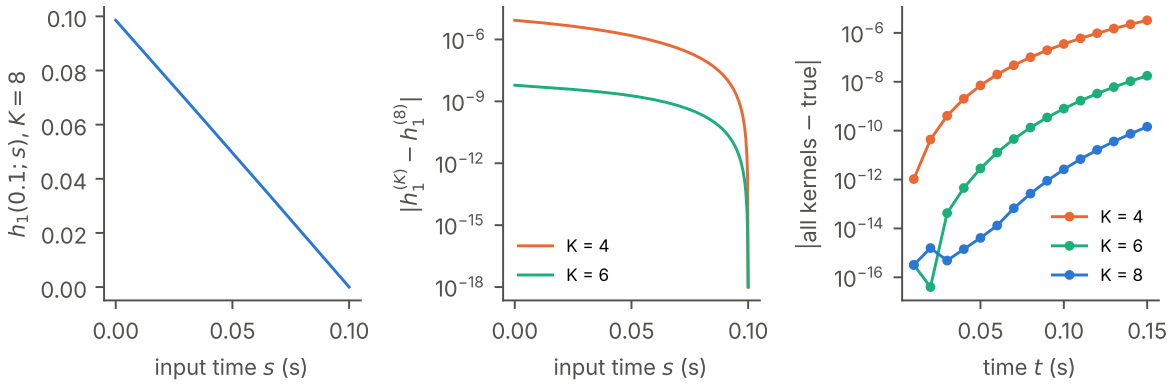



Figure 3: Left: first-order kernel $h_1(0.1; s)$ of the series truncated at $K = 8$. Middle: its difference from the kernels at $K = 4$ and $K = 6$. Right: difference between the response computed from all kernels of all orders and the true response of the oscillator.

it with directly; below, kernels from fitted coefficients are compared with the kernels of the exact series, not with responses of the system. The sum of the contributions of all orders, each computed from its kernel, should reproduce the true response. We solved (9) to high accuracy for the same input, at 15 times between 0.01 and 0.15 s. The largest difference between the sum over all kernels and the true response is $3.3 \times 10^{-6}$, $1.8 \times 10^{-8}$ and $1.4 \times 10^{-10}$ for $K = 4$, 6 and 8, against a largest response of 0.064 (Figure 3). The difference falls as $K$ grows, because it is the error of truncating the series. Figure 4 shows the second-order kernel over the whole square of input times. It is symmetric about the diagonal $s_1 = s_2$, and has a corner along it, where the order of the two input times changes.

### 6.1 Kernels from fitted coefficients

The kernels are sums of coefficients, so they inherit the imprecision of the fitted coefficients of long words. With 1 % noise and word selection, the first-order kernel $h_1(0.1; s)$ of the driven oscillator has a largest error of 1.4 % of the largest value of the kernel at $K = 4$ and 0.6 % at $K = 6$ (medians over 50 noise draws). With all words fitted by least squares the errors are 5 % and 4682 %: at $K = 6$ the errors of the long words swamp the kernel. The second-order kernel could not be recovered. At $K = 4$ the truncated series has no second-order term at all, yet the selected fit gives a second-order kernel whose largest value has a median of $5.7 \times 10^{-2}$ over the noise draws, 4500 times the largest value of the second-order kernel of the series at $K = 6$. At $K = 6$ the median error of the selected fit is $1.3 \times 10^{-5}$, the whole size of the kernel, as it is when the fit keeps no word with two input letters and so has no second-order kernel at all; with all words fitted the error is $4.3 \times 10^{7}$ times the largest value of the kernel. For the test inputs the second-order part of the series is at most $1.5 \times 10^{-8}$ over 0.1-0.15 s, far below the noise, so the data carry almost no information about it.

### 6.2 Kernels without a letter for time

Without a drift letter, every word of length $n$ has $n$ channel letters and no runs of time letters between them. Its integral is

$$I_{(a_1,\dots,a_n)}(t) = \int_{0<s_1<\cdots<s_n<t} u_{a_1}(s_1) \cdots u_{a_n}(s_n)\, ds,$$

so the order-$n$ kernel for the channels $a_1, \dots, a_n$, taken in that time order, is the constant $c_{(a_1,\dots,a_n)}$ on the ordered region. The kernels of a path signature therefore carry no time dependence of their own: all of it comes from the inputs. Their symmetric form, defined as in

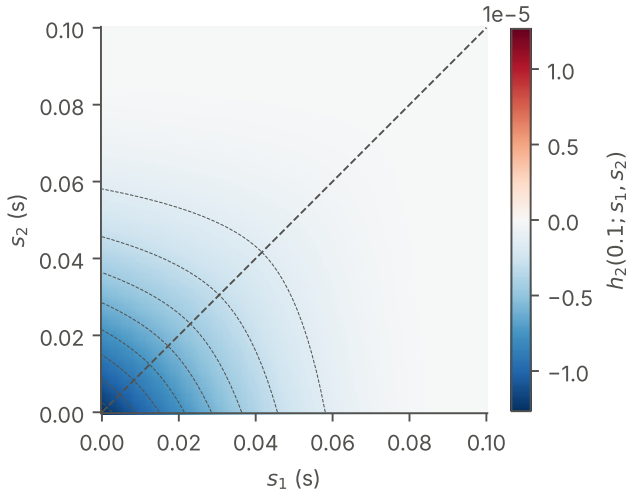



Figure 4: Symmetric second-order kernel $h_2(0.1; s_1, s_2)$ of the driven Duffing series at $K = 8$, for all $s_1, s_2$ in $[0, 0.1]$ s, with contours. The dashed line is the diagonal $s_1 = s_2$, about which the kernel is symmetric.

Proposition 4 over the whole cube, is piecewise constant, with one value for each order of the channels in time.

# 7 The Lyndon basis

## 7.1 Lyndon words and Lie polynomials

Many words carry overlapping information. For example, the integrals of $(0, 1)$ and $(1, 0)$ in Example 1 add up to $t\, I_{(1)}(t)$, the product of the integrals of $(0)$ and $(1)$. Lyndon words select a set of words from which all the others can be built by such products. They do not remove any word from a linear model: the integral of a non-Lyndon word is a polynomial in the integrals of Lyndon words, not a weighted sum of them.

Order the letters $0 < 1 < \cdots < m$, and compare words as in a dictionary. A rotation of a word moves some of its first letters to the end; the rotations of $(0, 0, 1)$ are $(0, 1, 0)$ and $(1, 0, 0)$. A word is a Lyndon word if it comes strictly before each of its rotations. So $(0, 1)$ is a Lyndon word, and $(1, 0)$ and $(0, 0)$ are not. With one input, the Lyndon words up to length 4 are

$$(0),\ (1),\ (0,1),\ (0,0,1),\ (0,1,1),\ (0,0,0,1),\ (0,0,1,1),\ (0,1,1,1).$$

There are 2, 1, 2, 3, 6, 9, 18 and 30 of lengths 1 to 8, against $2^L$ words of length $L$.

Each Lyndon word $l$ of length at least 2 is split as $l = uv$, where $v$ is its longest proper ending that is itself a Lyndon word; then $u$ is a Lyndon word too [4]. The standard bracketing of $l$ is the combination of words $P_l = P_u P_v - P_v P_u$, with $P_a = a$ for a single letter. For example,

$$P_{(0,1)} = (0,1) - (1,0), \qquad P_{(0,0,1)} = (0,0,1) - 2\,(0,1,0) + (1,0,0).$$

Each $P_l$ is built from commutators, and is called a Lie polynomial. The $P_l$, for $l$ a Lyndon word, form a basis of all Lie polynomials [28].

The Lie polynomials have a meaning for the system (1). Extend the coefficients to combinations of words by linearity, so that $c_{P_{(0,1)}v} = c_{(0,1)v} - c_{(1,0)v}$. This is the difference between applying the drift and then the input, and applying the input and then the drift. By (3) it equals $\big((L_{f_0}L_{f_1} - L_{f_1}L_{f_0})L_v h\big)(x_0)$. The commutator of two Lie derivatives is the Lie derivative along

a third vector field, the Lie bracket $[f_0, f_1]$, defined so that $L_{[f,g]} = L_f L_g - L_g L_f$: a direction in which the state can be moved by alternating drift and input. In the same way, for every Lyndon word,

$$c_{P_l v} = \big(L_{F_l} L_v h\big)(x_0), \tag{11}$$

where $F_l$ is the bracket of the vector fields $f_0, \ldots, f_m$ named by $P_l$.

### 7.2 The Lie–Hankel matrix

The Lie–Hankel matrix of a series has one row for each Lyndon word $l$ and one column for each word $v$, with entry $c_{P_l v}$. The Lie–Hankel matrix of a series truncated at order $K$ uses the same convention as the staircase matrix: rows of length 1 to $K$, columns of length 0 to $K$, and entries with $|l| + |v| > K$ set to zero, because those are coefficients of the truncated series beyond order $K$. Every row is a Lie polynomial with no constant term, so the empty-word coefficient $c_{()}$, and with it the starting value, never enters either matrix.

**Proposition 5.** *For the (untruncated) series of a system* (1) *with $n$ states, the Lie–Hankel matrix has rank at most $n$.*

*Proof.* By (11), $c_{P_l v} = \nabla(L_v h)(x_0) \cdot F_l(x_0)$. Each entry is the product of a row vector that depends only on $v$ and a column vector that depends only on $l$, both of length $n$. □

Fliess showed more [10]: a series has a realisation of the form (1) with analytic vector fields, near $x_0$, if and only if the rank of its Lie–Hankel matrix (its Lie rank) is finite, under a convergence condition that every polynomial series meets; the smallest such realisation then has as many states as the Lie rank. This is the nonlinear counterpart of the result for the ordinary Hankel matrix and bilinear systems. Realisations of series that are polynomials, and nilpotent approximations, were studied computationally by Hoang Ngoc Minh, Jacob and Oussous [16] and by Oussous [23, 24]; the existence and uniqueness of minimal nonlinear realisations go back to Sussmann [32].

For the full driven Duffing series the Lie rank is 2. This follows from Proposition 5, which bounds it by 2, and from its smallest block, rows $(0), (1)$ and columns $(), (0), (1)$, which already has rank 2; the blocks computed from exact coefficients up to length 10 confirm it. A bilinear system is linear in its state, so a bilinear realisation needs extra states to represent the cubic term, and the ordinary Hankel rank grows with the order.

Table 12 gives the rank of the Lie–Hankel matrix of the truncated series, with its zeros beyond order $K$. It grows with $K$, more slowly than the staircase rank. The truncated series is a polynomial, not the series of the oscillator, and by Fliess's result its smallest nonlinear realisation needs this many states. The staircase rank, the size of its smallest bilinear realisation, is larger at every order. The Lie–Hankel matrix does not by itself give a bilinear realisation; its value here is to separate the number of states of a nonlinear model from the size of a bilinear approximation.

The same holds without a drift. The full driftless Duffing series has Lie rank 2, its number of states, while the Lie–Hankel ranks of its truncations are 2, 3, 4, 5, 6, 7, 8 for $K = 2$ to 8, against staircase ranks of 3, 4, 5, 6, 7, 8, 9. For the Heisenberg integrator, whose series is finite, the Lie rank is 3, its number of states, against a minimal bilinear realisation with 4 states.

From fitted coefficients the Lie–Hankel matrix has a sharp gap in its singular values, but at the wrong place. For the exact truncated series its rank is 4 at $K = 4$ and 7 at $K = 6$. With all words fitted by least squares it is 6 and 11, with or without noise, which is the largest possible value for any series truncated at that order (Table 10): the errors in the coefficients of long words fill every direction that truncation allows. With word selection it is still 6 and 11 without noise, where the extra words take up the errors of truncation and sampling, and at 1 % noise it is 6 at $K = 4$ and 3 at $K = 6$, where only 4 words are kept (one noise draw each, with inputs

Table 12: Rank of the Lie–Hankel matrix of the driven Duffing series truncated at order $K$ (rows: Lyndon words up to length $K$; columns: all words up to length $K$; entries beyond order $K$ zero), and the staircase rank, the minimal bilinear size, from Table 1.

| $K$ | Lyndon rows | columns | Lie–Hankel rank | staircase rank |
|---|---|---|---|---|
| 2 | 3 | 7 | 2 | 3 |
| 3 | 5 | 15 | 3 | 4 |
| 4 | 8 | 31 | 4 | 5 |
| 5 | 14 | 63 | 6 | 7 |
| 6 | 23 | 127 | 7 | 9 |
| 7 | 41 | 255 | 8 | 12 |
| 8 | 71 | 511 | 9 | 16 |

Table 13: Regression in four bases at $K = 6$, 0.15 s. $R^2$ on the test trials without noise (one fit) and at 1 % noise (standard deviation 1 % of that of the response; median over 50 draws), with word selection. The Lyndon products span the same model as all words: fitted by least squares on all words, the predictions in the two bases agree to $1.0 \times 10^{-13}$; they were not fitted with selection, which would choose a different set. Condition numbers are for the full bases, columns scaled to unit length.

| basis | coefficients | condition number | $R^2$, no noise | $R^2$, 1 % noise |
|---|---|---|---|---|
| all words | 126 | $1.8 \times 10^6$ | 1.0000 | 1.0000 |
| Lyndon products $R_w$ | 126 | $1.9 \times 10^8$ | not fitted | not fitted |
| Lyndon words only | 23 | $8.6 \times 10^3$ | 0.9970 | 0.9970 |
| log-signature, linear | 23 | $3.4 \times 10^2$ | 0.9970 | 0.9970 |

drawn for this check). The fitted rank is at the ceiling or below the exact value, and in neither case does it count the states of the system.

### 7.3 Lyndon words in the regression

Products of iterated integrals are sums of iterated integrals: $I_u I_v$ is the sum of $I_w$ over all the ways $w$ of interleaving the letters of $u$ and $v$ while keeping the order within each [26]. For example, $I_{(0)}I_{(1)} = I_{(0,1)} + I_{(1,0)}$ and $I_{(1)}^2 = 2I_{(1,1)}$. Every word $w$ can be written in one way as a sequence of Lyndon words that never increase, $w = l_1^{n_1} \cdots l_k^{n_k}$ with $l_1 > \cdots > l_k$ [4]. The products

$$R_w = \frac{I_{l_1}^{n_1} \cdots I_{l_k}^{n_k}}{n_1! \cdots n_k!}$$

form a second basis, in which every iterated integral is a polynomial in the integrals of Lyndon words [25, 28]. For example, $(1, 0)$ splits as $(1)(0)$, and $R_{(1,0)} = I_{(1)}I_{(0)} = I_{(0,1)} + I_{(1,0)}$.

Lyndon words have been used in two other ways for regression on iterated integrals. One uses only the Lyndon words as regressors. The other uses the log-signature: the logarithm of the series $\sum_w I_w\, w$, taken in the algebra of words, which is a Lie polynomial and so is fixed by one coordinate for each Lyndon word, and reads the response as a weighted sum of those coordinates. Table 13 compares the four choices at $K = 6$.

Fitting the response to all the $R_w$ instead of all the $I_w$ is the same model in different coordinates, because each set is an invertible combination of the other. With all words fitted by least squares, the predictions of the two fits on the test trials agree to within $1.0 \times 10^{-13}$. With word selection the two bases would choose different sets of words, so the Lyndon products were not fitted with selection. The condition number is worse: $1.9 \times 10^8$ in the Lyndon basis, against $1.8 \times 10^6$ in the word basis. This is general: the uncertainty of any coefficient, or of any kernel,

does not depend on the coordinates in which the model is written, and a change of coordinates only moves the ill-conditioning into the map between them.

The other two are different, smaller models with 23 instead of 126 coefficients. Products of Lyndon integrals are not weighted sums of them; $I_{(1,1)} = \frac{1}{2} I_{(1)}^2$ is a square, and no weighted sum of Lyndon integrals can produce it. The same holds for the log-signature coordinates. Both models are much better conditioned, because they have far fewer coefficients, and both lose accuracy. With word selection in every case, their $R^2$ on the test trials stays well below that of the full series both without noise and at 1 % noise (Table 13): their error is mainly one of the model, not of noise. The useful role of Lyndon words is therefore the Lie–Hankel matrix of the previous section, not a basis for the fit.

# 8 Practical lessons

1. Compute iterated integrals exactly when the input is a formula, and with iisignature on the sampled path when it is data. Report which was used and the sampling error, which for long words is far larger than for short ones.

2. Build the Hankel matrix of a truncated series with rows and columns of all words up to length $K$, and zeros wherever the combined word is longer than $K$. Its rank gives the minimal realisation, which is nilpotent.

3. The graded construction stays nilpotent when it is reduced, but it is larger, and in the examples its guarantee did not change any prediction within the window.

4. Shift each window to start at zero and fit a free offset for each window.

5. Judge a fit by its $R^2$ on data not used for fitting, and use enough trials for the number of coefficients.

6. Many coefficients are zero because of the structure of the system. Select words jointly over all windows that share a starting state, choose the number of words on separate validation data, and refit the chosen words without a penalty. Do not use ridge regression: it cannot set a coefficient to zero, and it shrinks the fitted coefficients. Selection does not find the exact set of words either. The words kept depend on the noise, the length of the data and how well the columns can be told apart, so report them.

7. Compare coefficients only on a common scale, such as $c_w A^n T^L / L!$, and only as relative errors. The individual coefficients could not be estimated from noisy data. The main reasons are that the terms of long words are small, the columns are nearly dependent, the words made only of the time letter are the same in every trial, and the integrals computed from samples are least accurate for long words; truncation adds a smaller error. The response could be fitted.

8. With all words fitted, the sizes of realisations of fitted series, and their Lie–Hankel ranks, sit at the largest values possible for the order. With word selection they follow the number of words kept: for the driven oscillator they reach the ceiling without noise, and they fall as the noise grows, to below the exact value at high noise. Compare any such size with the ceiling, and compare recordings only at the same noise level, length and selection rule.

9. The size of the smallest realisation belongs to the starting state. It can be smaller at special states, such as an equilibrium, so a recording that passes through different regimes can have a local rank that changes along it. Fitted sizes cannot show this unless they lie below the ceiling and are compared at the same noise level.

10. Validate kernels only through the full series: summed over all orders, they must reproduce the true response. Without a drift, the kernels are the coefficients themselves.

11. A truncated series, and so its realisation, describes the system near the starting state only: within a window of time with a drift, and without one only while the iterated integrals of the path, including the areas it sweeps, stay small. A finite series, such as that of the Heisenberg integrator, is the exception.

12. Adding time to the channels of a path signature gives the Chen–Fliess series with a letter for time, so everything here applies to such time-augmented signatures.

13. Lyndon words give the Lie–Hankel matrix, whose rank, for exact coefficients, counts the states of a nonlinear model; for fitted coefficients it does not. As a basis for the fit they give the same fit or a smaller, less accurate model.

## Code availability

The code that produces every number, table and figure of this tutorial is released with it as an open-source package under the MIT licence; its README explains how to use it.

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